



National University of Computer & Emerging Sciences, Karachi
Midterm Two (Fall-2023)



9th November, 2023 10:00 AM – 11:00 AM

Course Code: AI-3002	Course Name: Machine learning
Instructor Name: Dr. Muhammad Farrukh Shahid, Sania Urooj	
Student Roll No:	Section:

Instructions:

- Return the question paper and make sure to keep it inside your answer sheet.
- Read each question completely before answering it. There are **4 questions and 3 pages**.
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper.
- You are **not allowed to write** anything on the question paper (except your ID and group).

Time: 60 minutes.

Max Marks: 50 Points

Question No.1:

CLO 2 [pt = 5]

Identify whether the below statements are True or False. There will be a deduction by 0.5 for any wrong answer.

1. A deep neural network has many hidden layers, making it capable of learning complex and hierarchical features from data.
2. In backpropagation neural networks, information flows in one direction, from input to output layers, without any loops or cycles.
3. The coefficients learned in Logistic Regression indicate the impact of each feature on the log-odds of the target variable.
4. A small learning rate may lead to slow convergence in gradient descent, while a large learning rate may lead to overshooting the minimum
5. Gradient descent is guaranteed to find the global minimum of a cost function for any given problem.

Question No. 2.

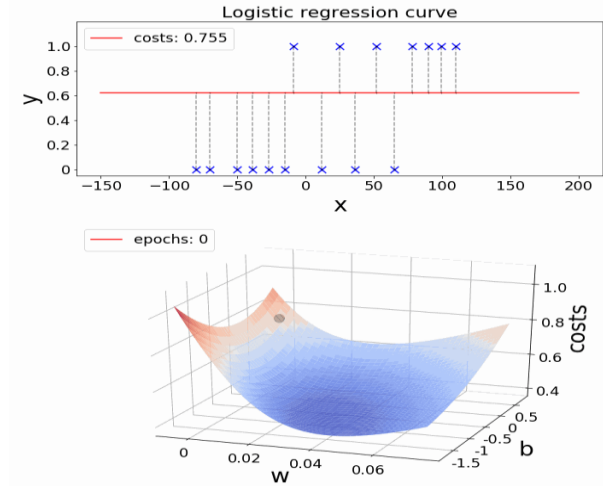
CLO 2 [pt = 6 + 6]

- a) Consider the following single neuron network having a sigmoid activation function. Initial weights are randomly initialized as [0.5 0.5]. Following data points are given.

1. [0.2 0.4]
2. [0.4 0.5]
3. [0.6 0.2]

Find the final updated weights.

- b) Given the Logistic regression learning problem as shown in the figure below. Do the following:
1. Describe in 4 bullet points the overall intuition behind the logistic regression.
 2. Explain the concept of the convex function in the context of the graph shown below.
 3. Develop a case study in which you demonstrate the need and motivation of using LR. Your case study should contain [The problem statement, dataset, LR, Expected outcome].



Question No. 3

CLO 2 [pt = 10 + 5]

- a) Find the best fitted line using estimated coefficients for the below given dataset in Table 1. Predict the bill of a person who uses 6 appliances in a month. Also, calculate the cost function.
- b) Imagine you have calculated the gradient of the cost function. Explain the role of this gradient in the gradient descent algorithm. After computing the gradient, how does gradient descent update the model parameters in the linear regression problem? Also, describe the role of learning rate in updating rules.

Table 1:

Number of appliances	Monthly Bill
2	80
3	120
4	150
5	180
8	210
9	280
10	340

Question No. 4

CLO 2 [pt = 6 + 12]

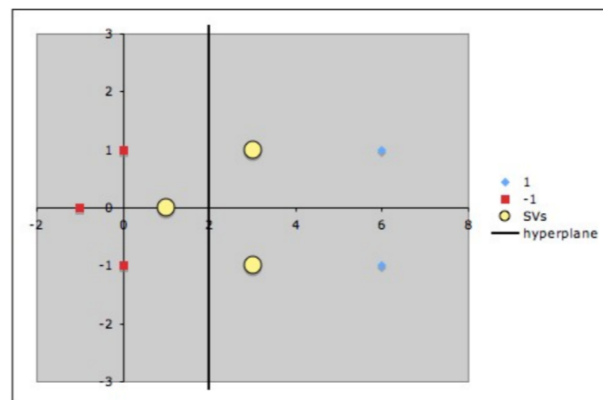
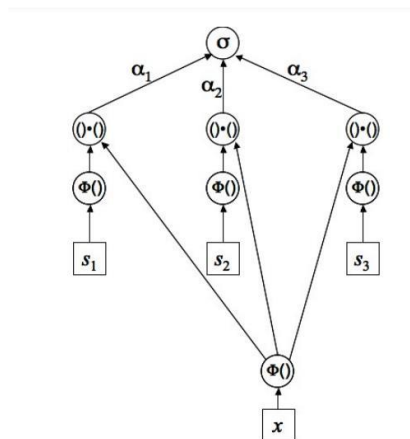
- a) Consider a binary classification problem with two inputs and the following labeled dataset for training. Plot the points on a 2D plane. Can we use hard SVM for this problem? Provide a brief justification to support your answer

Table 2: SVM Dataset

Label y	Data Point x (1), x (2)
1	-2,-2
1	-2,2
1	2,2
-1	1,1
-1	1,-1
-1	-1,1

- b) Given the graph of the SVM showing the discriminating hyperplane corresponding to the values of α_1 , and having $b = -2$ and $w = [1 \ 0]^T$. The values of α_2 and α_3 are 0.75 for both. What will be the value of α_1 such that we obtain the hyperplane as shown in the figure below. Assume $\Phi() = I$ (identity matrix) and use the following equation,

$$\tilde{w} = \sum_i \alpha_i \tilde{s}_i$$



Good Luck !